1.
You are on a cruise in the Southern Ocean and the science team has discovered a new hydrothermal system. Preliminary analysis indicates that, in the diluted plume, dFe = 10 nM and the pH is 7.5. Strong Fe-binding ligands are present at a concentration of 20 nM and have a equilibrium constant (K) of 10^{11}, relative to inorganic iron (Fe'). Based on other tracers, it is estimated that the waters in the plume are part of a water mass with a ventilation timescale of 50 years, which will outcrop in iron-limited waters with dFe ~0.1 nM.

Part 1
With the meager internet at your disposal, you manage to download a paper that estimates the rate of precipitation of iron oxy-hydroxides (FeOOH) based on lab experiments. The precipitation reaction follows the equation:

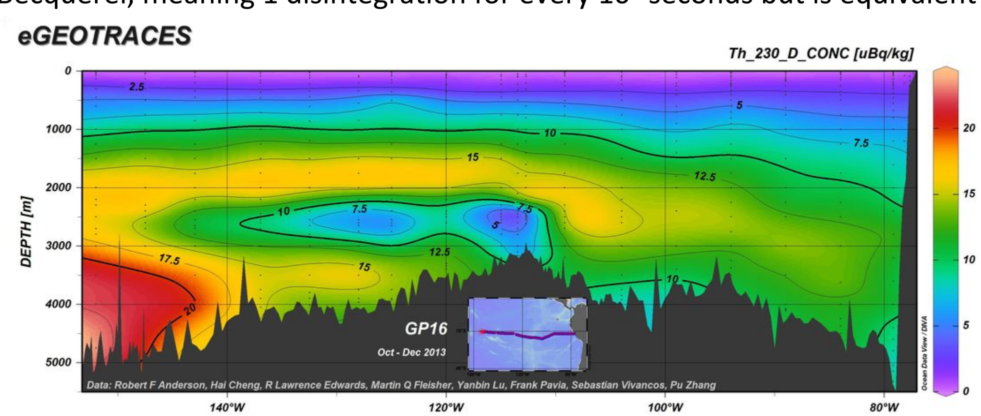
Precipitation Flux = $c*[Fe']*[OH^-]$

where c = 500 M^{-1} s^{-1}. If FeOOH precipitation is the only removal process in the plume, what is the turnover time of iron in the plume (i.e. what is its residence time)? How much will the hydrothermal Fe source impact the dFe concentration in the Southern Ocean surface waters?

Part 2
A year passes. An eruption has caused the vent chemistry to change and your colleague returns to re-characterize this site. The pH and ligand concentrations are the same as your team observed previously, but now plume water dFe = 100 nM. The team suggests that 10x more Fe will now reach the surface ocean. Are they right?

Part 3
During the last glacial maximum, atmospheric CO_2 levels were much lower than found today, with a larger storage of inorganic carbon in the deep ocean leading to lower pH. How would this affect Fe scavenging (precipitation rates) in this plume and Fe supply to the Southern Ocean?

Part 4
Explain the distribution of ^{230}Th in the South Pacific in 1-2 sentences. Note that uBq is a micro-Becquerel, meaning 1 disintegration for every 10^6 seconds but is equivalent to a concentration.



2. Does ocean acidification act as a positive, negative, or neutral feedback on climate change?